

```

In[78]:= (*Defining our coordinates*)coords = {t, r,  $\theta$ ,  $\phi$ };

(*Defining the metric*)
g = {{f[r], 0, 0, 0}, {0, h[r], 0, 0}, {0, 0,  $r^2$ , 0}, {0, 0, 0,  $r^2 \text{Sin}[\theta]^2$ }};

(*Defining the inverse metric*)
ginv = Simplify[Inverse[g]];

n = Length[coords];

(*Defining the Christoffel symbols  $\Gamma^{\rho}_{\mu\nu}$ *)
Gamma1 =
Table[1/2 Sum[ginv[[ $\rho$ ,  $\sigma$ ]] (D[g[[ $\sigma$ ,  $\nu$ ]], coords[[ $\mu$ ]] + D[g[[ $\sigma$ ,  $\mu$ ]], coords[[ $\nu$ ]] - D[g[[ $\mu$ ,  $\nu$ ]], coords[[ $\sigma$ ]]),
{ $\sigma$ , 1, n}], { $\rho$ , 1, n}, { $\mu$ , 1, n}, { $\nu$ , 1, n}];

(*Simplifying the results*)
Gamma2 = Simplify[Gamma1];

(*Printing the Christoffel symbols in table format*)
TableForm[Gamma2, TableHeadings -> {coords, coords}]

```

Out[84]//TableForm=

	t	r	θ	ϕ
t	0	$\frac{f'[r]}{2 f[r]}$	0	0
	$\frac{f'[r]}{2 f[r]}$	0	0	0
	0	0	0	0
	0	0	0	0
	0	0	0	0
r	$-\frac{f'[r]}{2 h[r]}$	0	0	0
	0	$\frac{h'[r]}{2 h[r]}$	0	0
	0	0	$-\frac{r}{h[r]}$	0
	0	0	0	$-\frac{r \text{Sin}[\theta]^2}{h[r]}$
	0	0	0	0
θ	0	0	$\frac{1}{r}$	0
	0	$\frac{1}{r}$	0	0
	0	0	0	$-\text{Cos}[\theta] \text{Sin}[\theta]$
	0	0	0	0
	0	0	0	$\frac{1}{r}$
ϕ	0	0	0	$\text{Cot}[\theta]$
	0	$\frac{1}{r}$	$\text{Cot}[\theta]$	0
	0	$\frac{1}{r}$	0	0